MPG No. 31.1.115 $\bar{4}3m$ [Type I, cubic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$M_{yzx} = \mathbb{T}_{0}^{(1)}$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & -M_{xxyy} & 0 & 0 & 0 \\ -M_{xxyy} & 0 & M_{xxyy} & 0 & 0 & 0 \\ M_{xxyy} & -M_{xxyy} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$M_{xxyy} = 4\mathbb{T}_{xyz}^{(1)}$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{yzyz} \end{bmatrix}$$

$M_{yzyz} = \mathbb{T}_{xyz}^{(2)}$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzyz} \end{bmatrix}$$

$M_{yzyz} = \mathbb{T}_{xyz}^{(3)}$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & M_{xxyy} & -M_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyy} & 0 & M_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xxyy} & -M_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 & -M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzyz} & 0 & 0 & -M_{yzyz} \end{bmatrix}$$

$M_{xxyy} = 4\mathbb{T}_{xyz}^{(1)}$
$M_{yzyz} = \mathbb{T}_{xyz}^{(2)}$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzyz} \end{bmatrix}$$

$M_{yzyz} = \mathbb{T}_{xyz}^{(3)}$

* Rank 4 tensor (t).